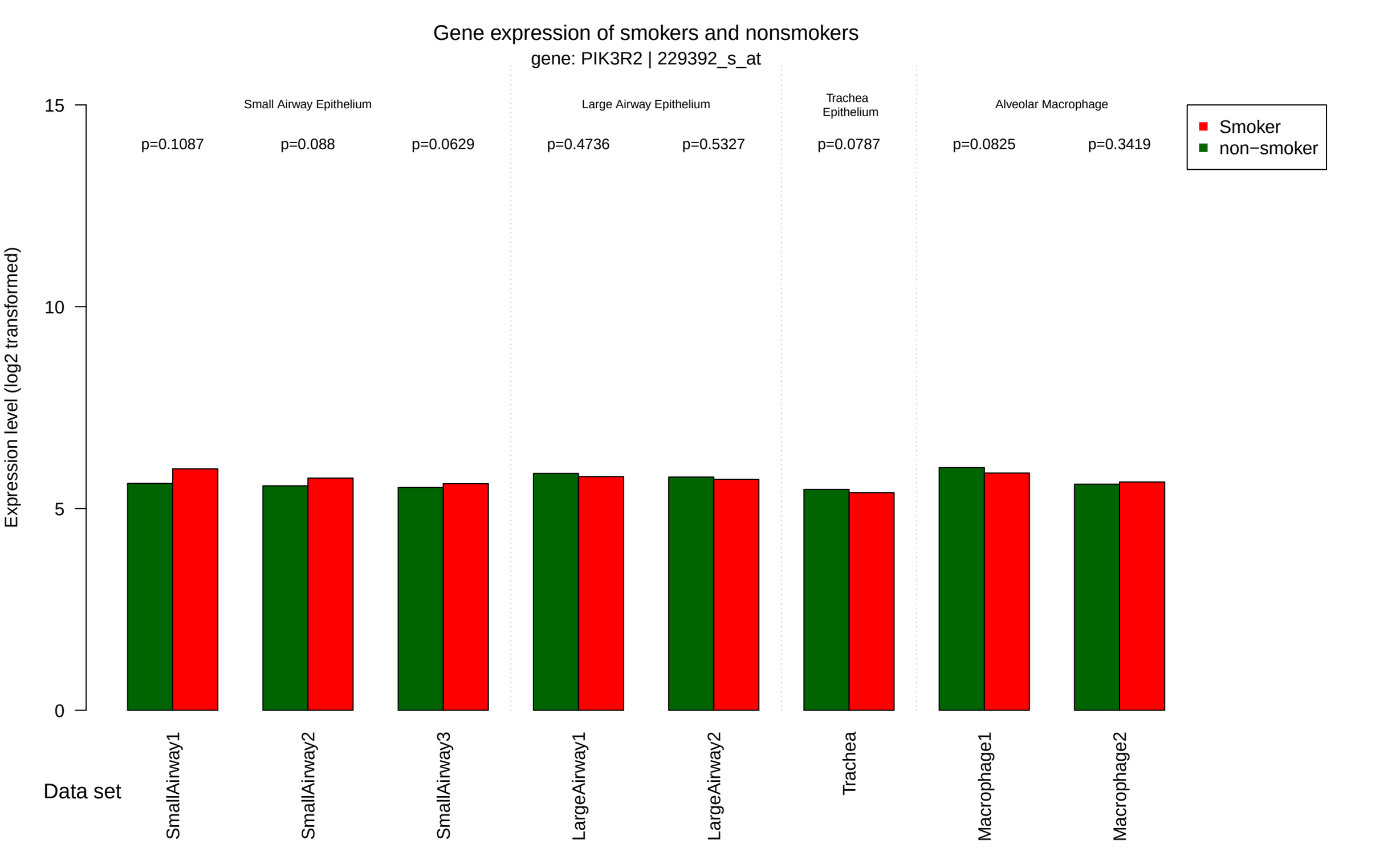
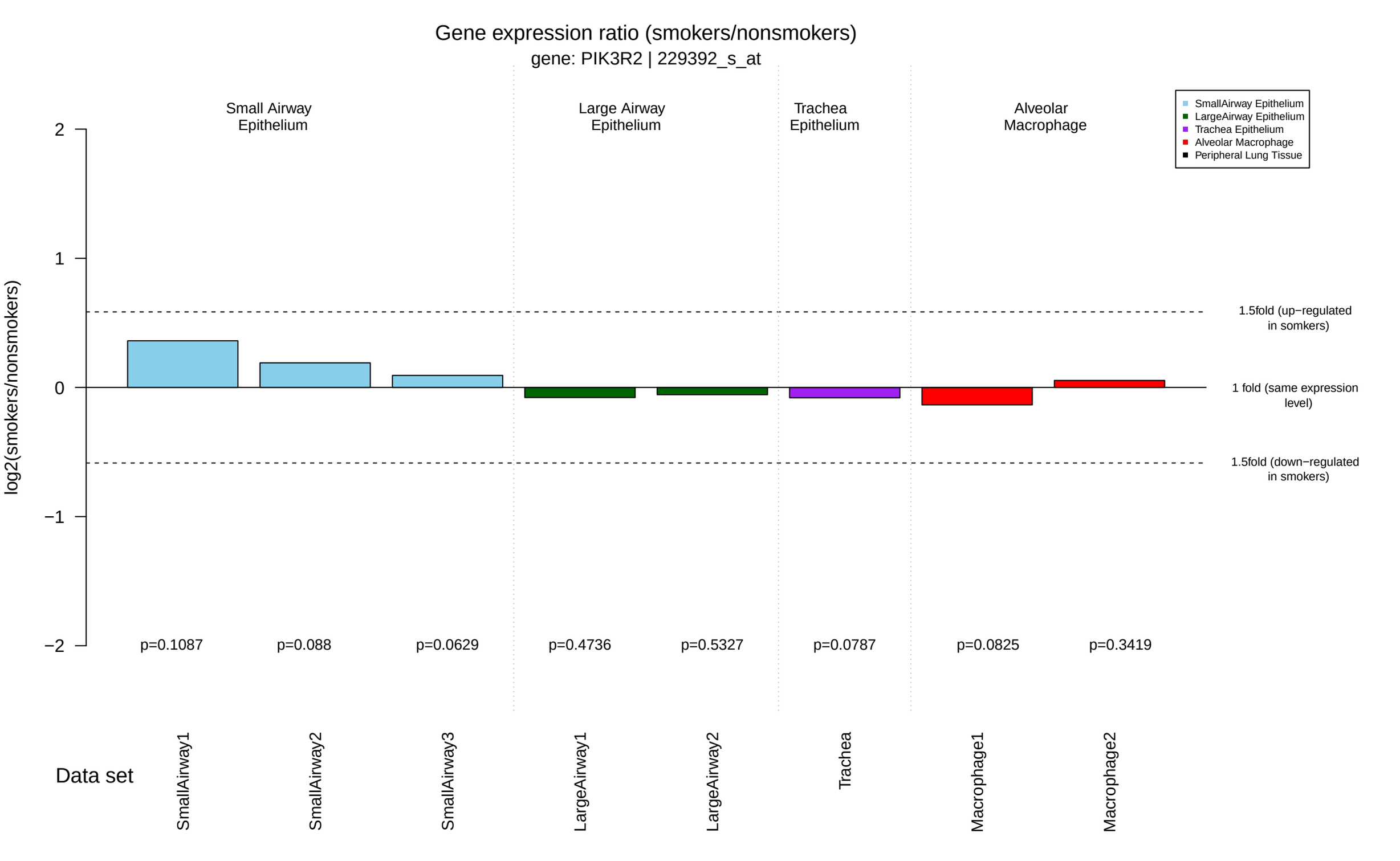
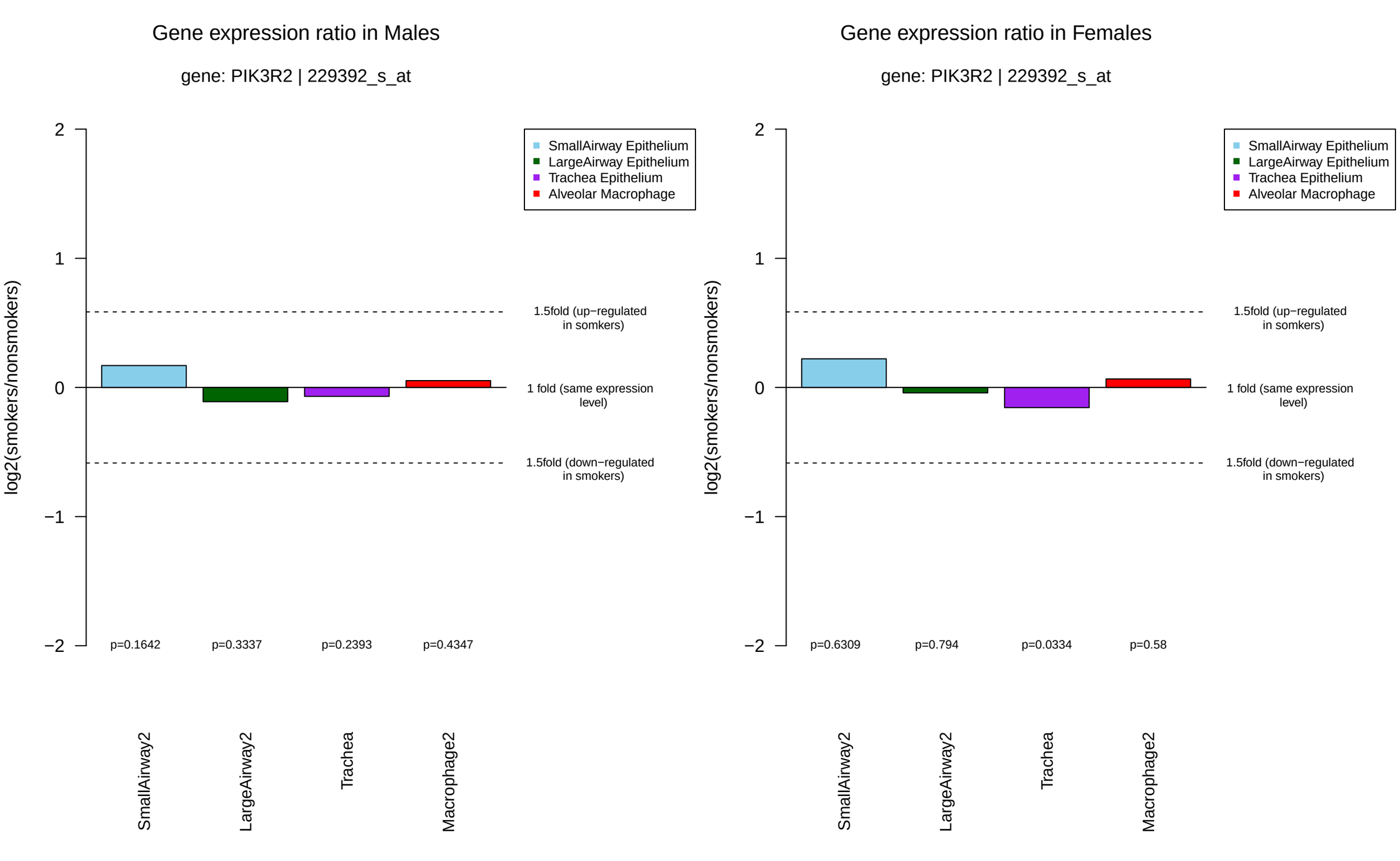
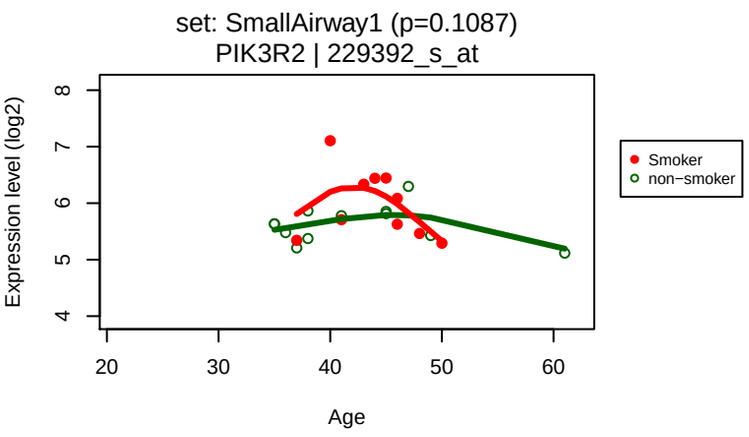
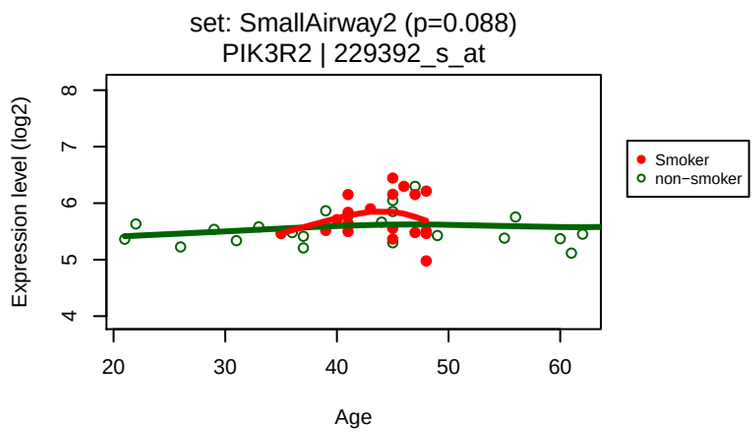
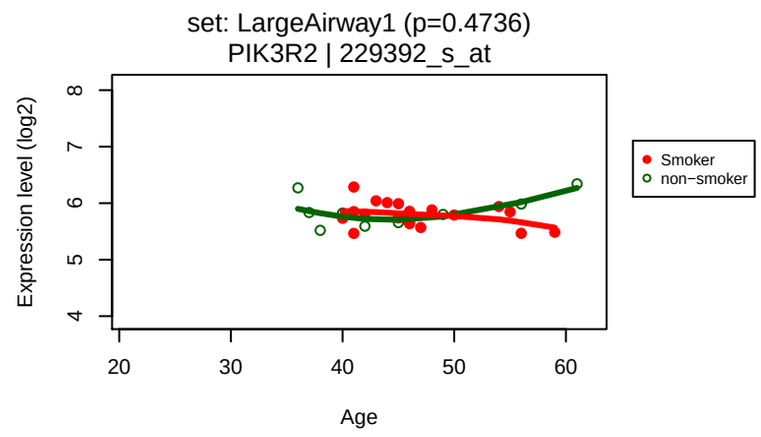

Gene expression of smokers and nonsmokers
gene: PIK3R2 | 229392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1087
p=0.088
p=0.0629
p=0.4736
p=0.5327
p=0.0787
p=0.0825
p=0.3419
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

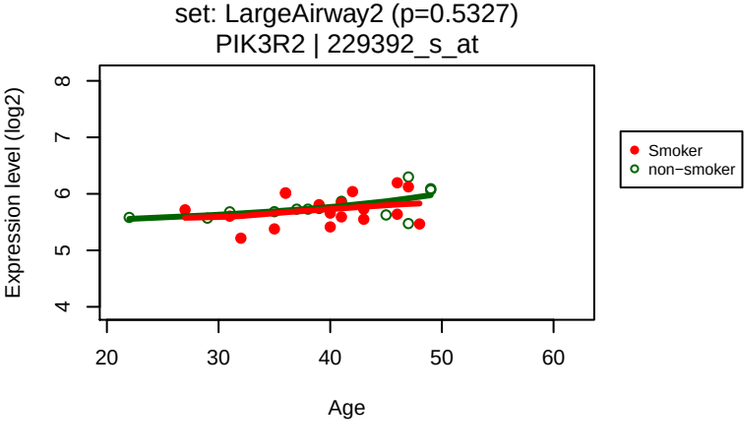

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R2 | 229392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1087
p=0.088
p=0.0629
p=0.4736
p=0.5327
p=0.0787
p=0.0825
p=0.3419
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

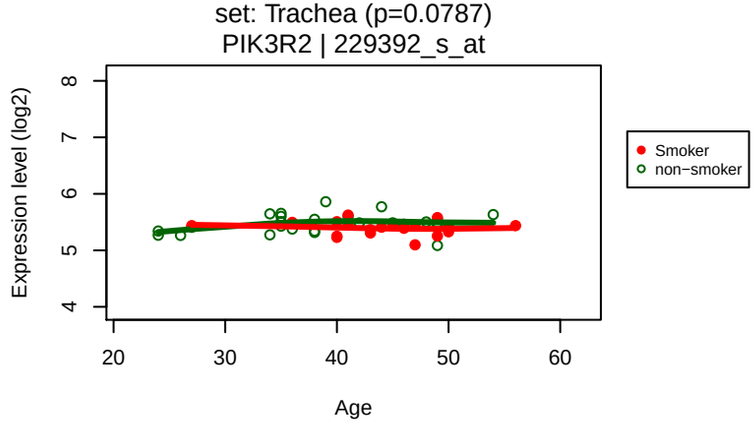

Gene expression ratio in Males
gene: PIK3R2 | 229392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1642
p=0.3337
p=0.2393
p=0.4347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R2 | 229392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6309
p=0.794
p=0.0334
p=0.58
SmallAirway2
LargeAirway2
Trachea
Macrophage2

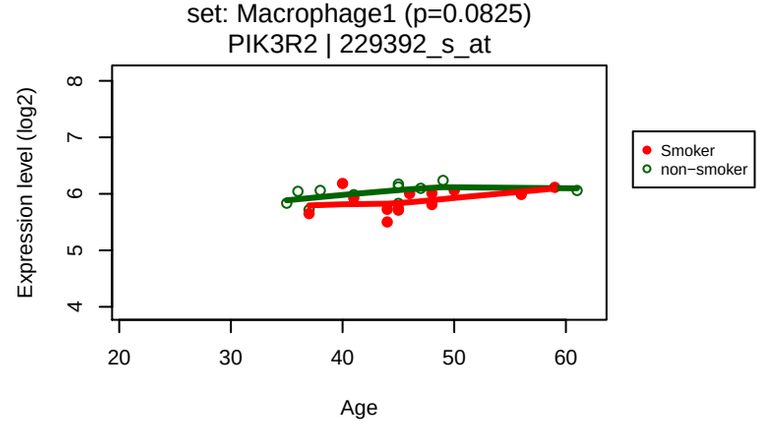

set: SmallAirway1 (p=0.1087)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker

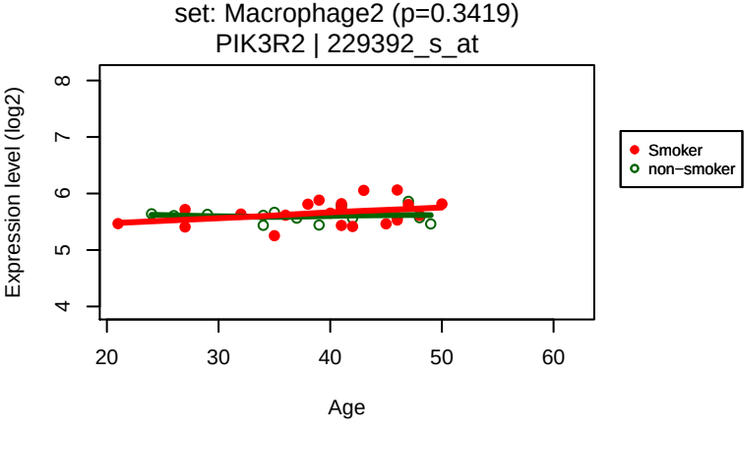

set: SmallAirway2 (p=0.088)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker

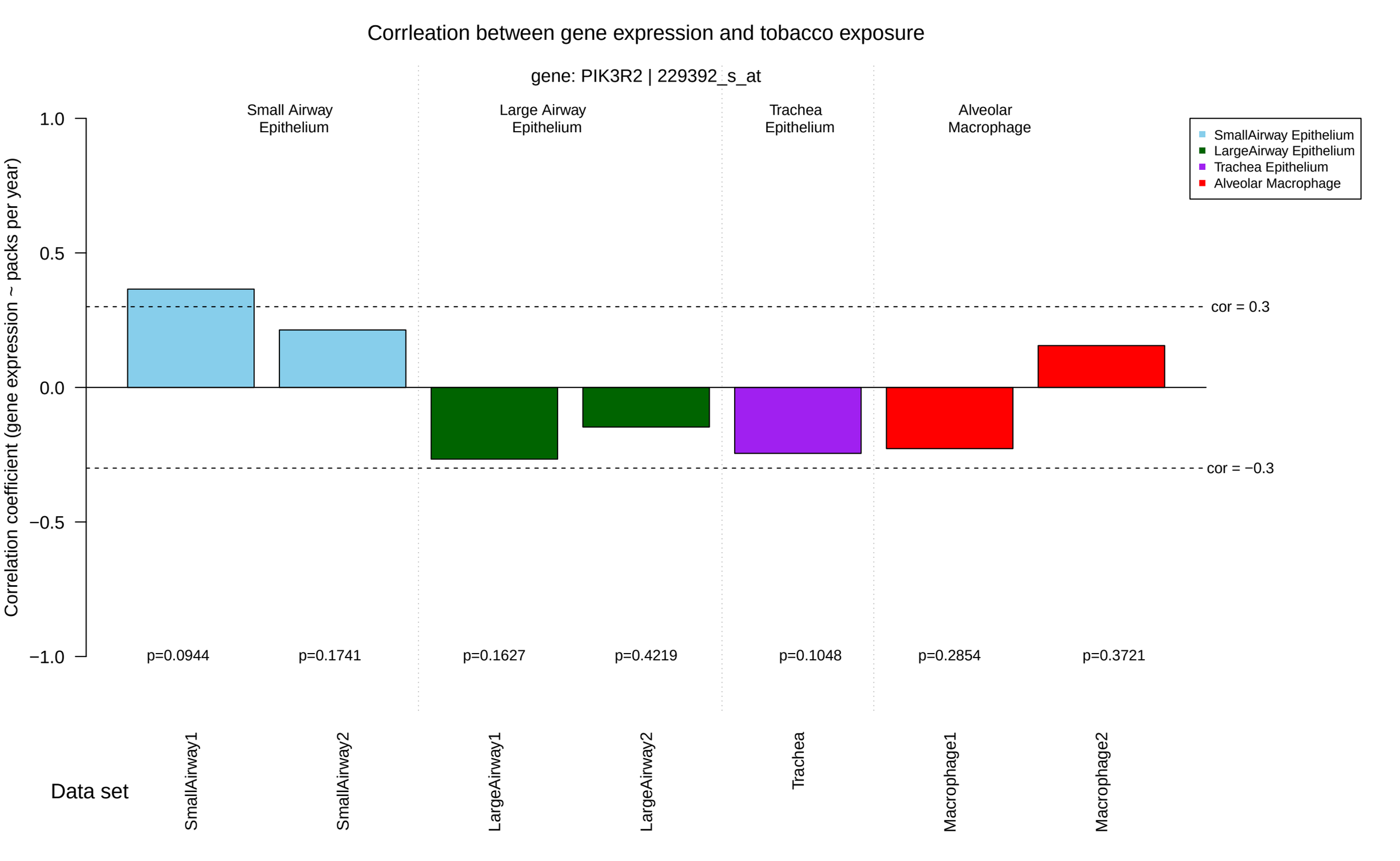

set: LargeAirway1 (p=0.4736)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5327)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0787)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0825)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3419)
PIK3R2 | 229392_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R2 | 229392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0944
p=0.1741
p=0.1627
p=0.4219
p=0.1048
p=0.2854
p=0.3721
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2